



Sydney Girls High School

2006
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Extension 2

This is a trial paper ONLY.
It does not necessarily
reflect the format or the
contents of the 2006 HSC
Examination Paper in this
subject.

General Instructions

- Reading Time - 5 mins
- Working time - 3 hours
- Attempt **ALL** questions
- **ALL** questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new sheet. Write on one side of the paper only.

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Candidate Number

Question 1

- a) If z is the complex number $-1+i$, indicate on the Argand Diagram, the points

$i) z, ii) \bar{z}, iii) iz, iv) \frac{1}{z}, v) z^2$ [5]

- b) If $z = \cos \theta + i \sin \theta$ and $w = \frac{dz}{d\theta}$, show that $z + iw = 0$. [2]

- c) Describe Geometrically the locus in the Argand Diagram represented by z if $z\bar{z} - 6(z + \bar{z}) = 45$. [2]

- d) Consider the equation $z^4 + 1 = 0$

- i) Find the four complex roots expressing them in the form $a+ib$
ii) If the roots are plotted on an Argand Diagram, find the area of the figure those roots form. [3]

- e) The interval AB where A is (2,3) and B(4,5) is rotated 60° about A in an anticlockwise direction to AB'. Find the co-ordinates of the point B' [3]

Question 2.

- a) Find the following definite integrals:

i) $\int_0^{\frac{1}{2}} \frac{dx}{1+9x^2}$

ii) $\int_0^{\frac{1}{2}} \sin^{-1} x \cdot dx$ [9]

iii) $\int_0^1 \frac{dx}{(x+1)(x+2)^2}$

b) Find $\int \frac{\sqrt{x^2-4}}{x^2} \cdot dx$ [3]

c) Let $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \cdot dx$, where n is an integer.

i) Show that $I_n = \frac{1}{n-1} - I_{n-2}$ [3]

ii) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^7 x \cdot dx$

Question 3:

a) If α, β, γ are the roots of the polynomial $3x^3 - 6x^2 - x + 1 = 0$

i) Find the value of $(\alpha-1)(\beta-1)(\gamma-1)$

ii) Hence or otherwise find the value of $(\alpha + \beta - \gamma)(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)$

[6]

b) The polynomial $ax^n + bx^{n-1} - 1$ where n is an even positive integer is divisible by $(x+1)^2$.

Show that $a = 1 - n$ and $b = -n$

[5]

c) Show that the product of 4 consecutive numbers is always one less than a perfect square.

[4]

Question 4:

a) Sketch the following curves showing all the important features of those curves

i) $y = (3+x)(1+x)^3(1-x)(3-x)^2$

ii) $y = \frac{x^2(x-3)}{(x-2)^2}$

iii) $4y^2 = x^2 - 4x$

[9]

b) Consider the function $f(x) = 4 - x^2$, $-2 \leq x \leq 2$ and sketch the following curves

i) $y = \sqrt{f(x)}$

ii) $y = \log_e \{f(x)\}$

iii) $y = 2^{f(x)}$

[6]

Question 5:

a) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$

i) Determine the eccentricity

[1]

ii) Find the foci and equation of the directrices

[1]

iii) Sketch the ellipse

[1]

iv) Find the equation of the tangent at the point $P(3 \cos \theta, 5 \sin \theta)$

[3]

v) Determine the equation of the tangent at $P(3 \cos \theta, 5 \sin \theta)$ in terms of the gradient m .

[2]

vi) Write down the equations of the tangents to $\frac{x^2}{9} + \frac{y^2}{25} = 1$ with gradient 2.

[1]

b) i) Find the equation of the tangent to $\frac{x^2}{8} - \frac{y^2}{4} = 1$ at the point (x_1, y_1)

[2]

ii) Hence write down the equation of the chord of contact to $\frac{x^2}{8} - \frac{y^2}{4} = 1$ from the point (x_0, y_0)

[1]

iii) If the equation of the chord of contact to $\frac{x^2}{8} - \frac{y^2}{4} = 1$ from the point $T(x_0, y_0)$ is

$$2x - 3y - 5 = 0, \text{ find the co-ordinates of point } T.$$

[3]

Question 6:

a) i) Find the exact value of $\tan 75^\circ$

[1]

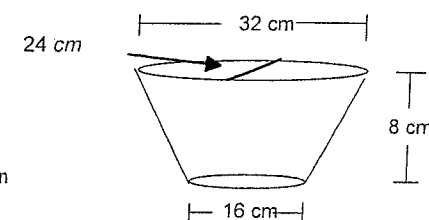
ii) Last month in the USA a trainee pilot stalled the engine of his plane and it nosedived, hitting the ground at an angle of 75° (being lucky enough to have that fall broken by a tree). If he was traveling in a horizontal direction at 108 km/hr when he stalled and the future motion of the plane was that of a projectile, how high was the plane when it stalled?

[3]

b) The area enclosed by a triangle with co-ordinates $A(1,1)$, $B(2,2)$ and $C(3,1)$ is rotated about the Y axis. Use the method of cylindrical shells to find the volume that is formed.

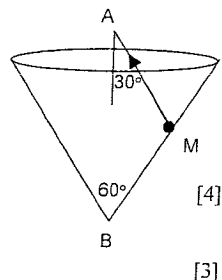
[5]

c) Maisie's birthday cake is in the shape of an ellipse with axes of 32 cm and 24 cm at one end and a circle of radius 8 cm at the other end. If the cake is 8 cm high, find the volume of the cake if each cross section taken perpendicular to the height is an ellipse



Question 7:

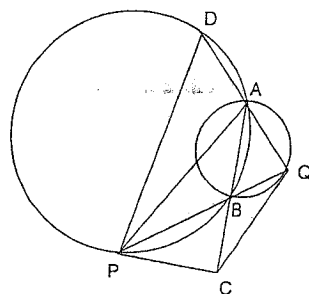
a) A mass M of 1 kg is attached to a string of length 4 metres at a point A directly above the vertex of a cone, with semi vertical angle of 30° . The string also makes an angle of 30° with the vertical.



- If the mass M rotates at 2 rad/sec, find the tension in the string and the normal force exerted by the side of the cone on the mass
- How fast (in rad/sec) should the mass be rotated in the tension (T) and the normal force (F) are to be equal in magnitude.

b) Two circles intersect at A and B . AB is produced to a point C , such that when tangents CP and CQ are drawn, PBQ is a straight line.

- Show that $CP = CQ$
- Show that $APCQ$ is a cyclic quadrilateral
- If QA is produced to meet the larger circle, at D , show that PB bisects $\angle CPD$.



Marks: i) [2], ii) [3], iii) [3]

Question 8:

a) If $xe^{-x} = k$ has two solutions, find the range of values of k

[4]

b) If the polynomial $x^4 + qx^2 + rx + s = 0$ has roots $\alpha, \beta, \gamma, \delta$ show that the value of the constant term in the polynomial with roots $1 - \alpha^2, 1 - \beta^2, 1 - \gamma^2, 1 - \delta^2$ is $(q + s + 1)^2 - r^2$

[6]

c) Evaluate $\int_0^4 \sqrt{2 - \sqrt{x}} dx$

[5]

.....end of paper.....

$$1a) z = -1 - i$$

$$\bar{z} = -1 + i$$

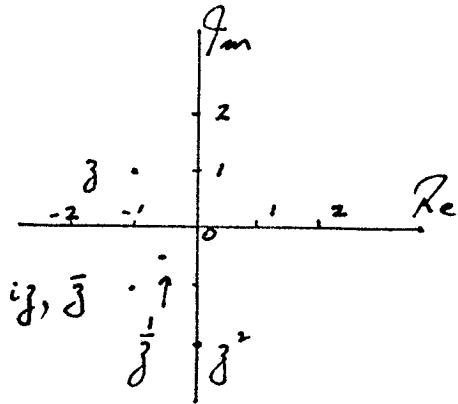
$$iz = -i - 1 = -1 - i$$

$$\frac{1}{z} = \frac{1}{-1-i} \cdot \frac{-1+i}{-1+i}$$

$$= \frac{-1+i}{2}$$

$$z^2 = (-1-i)^2$$

$$= -2i$$



$$b) z = \cos \theta + i \sin \theta$$

$$\frac{dz}{d\theta} = -\sin \theta + i \cos \theta = w$$

$$i w = -i \sin \theta - \cos \theta$$

$$\therefore z + i w = 0$$

$$c) z \bar{z} - 6(z + \bar{z}) = 45$$

$$\text{Let } z = x + iy$$

$$\therefore x^2 + y^2 - 12x = 45$$

$$(x-6)^2 + y^2 = 81$$

Circle, centre (6, 0), radius 9

$$d) z^4 + 1 = 0$$

$$z^4 = -1$$

$$\text{Let } z = \cos \theta + i \sin \theta$$

$$\therefore (\cos \theta + i \sin \theta)^4 = -1$$

$$\therefore \cos 4\theta + i \sin 4\theta = -1 \quad (\text{de Moivre's thm})$$

$$\cos 4\theta = -1$$

$$4\theta = \pi, 3\pi, 5\pi, 7\pi$$

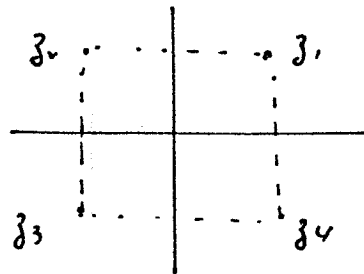
$$\theta = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$$

$$\therefore z_1 = \cos \pi/4 + i \sin \pi/4 = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$$z_2 = \cos 3\pi/4 + i \sin 3\pi/4 = -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$$z_3 = \cos 5\pi/4 + i \sin 5\pi/4 = -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$$

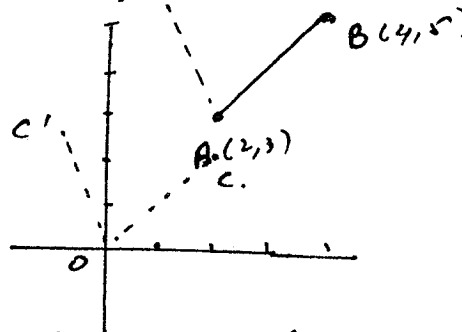
$$z_4 = \cos 7\pi/4 + i \sin 7\pi/4 = \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$$



$$A = z^2$$

$$= \left(\frac{2}{\sqrt{3}}\right)^2$$

$$\text{Then } = 2 \text{ sq units}$$



Translate AB to DC
- moving (-2, -3) places

Consider the interval OC
where O is (0, 0) & C is (2, 2)
Rotate C by 60° to C'

$$C' = (2+2i)(\cos 60 + i \sin 60)$$

$$= (2+2i)\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= 1 + i\sqrt{3} + i - \sqrt{3}$$

$$= (1-\sqrt{3}) + i(1+\sqrt{3})$$

$$\therefore B' \text{ is } (1-\sqrt{3}+2), i(1+\sqrt{3}+3)$$

$$= \{3-\sqrt{3}, i(4+\sqrt{3})\}$$

Q2. a)

$$i) \int_0^{\pi/3} \frac{dx}{1+\tan^2 x} = \left[\frac{1}{3} \tan^{-1}(3x) \right]_0^{\pi/3}$$

$$= \frac{1}{3} \tan^{-1}(1) - 0$$

$$= \pi/12$$

$$ii) \int_0^{\pi/2} \sin^{-1} x \, dx$$

$$\text{Let } u = \sin^{-1} x \quad dv = dx$$

$$du = \frac{dx}{\sqrt{1-x^2}} \quad v = x$$

$$\therefore I = [x \sin^{-1} x]_0^{\pi/2} - \int_0^{\pi/2} \frac{x}{\sqrt{1-x^2}} \, dx$$

$$= \left[\frac{1}{2} \sin^{-1} \left(\frac{1}{2}\right) + \left[\sqrt{1-x^2} \right]_0^{\pi/2} \right]$$

$$= \frac{1}{2} \cdot \frac{\pi}{6} + \sqrt{3/4} - 1$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$$

5(i) Let the consecutive numbers be

$$x-1, x, x+1, x(x+2)$$

Show $(x-1)x(x+1)(x+2)+1 = K^2$

$$\begin{aligned} \text{LHS} &= (x^3-x)(x+2)+1 \\ &= x^4+2x^3-x^2-2x+1 \end{aligned}$$

$$\begin{aligned} \text{Then } (x^2+x-1)^2 &= x^4+2x^3-2x^2-2x+1 \\ &= x^4+2x^3-x^2-2x+1 \end{aligned}$$

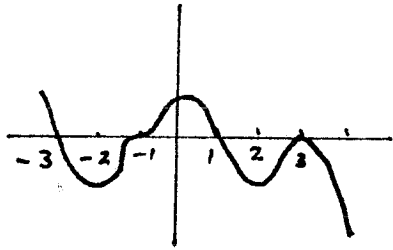
$$\begin{aligned} \therefore (x-1)x(x+1)(x+2)+1 \\ &= (x^2+x-1)^2 \end{aligned}$$

$$\text{OR } (x-1)x(x+1)(x+2) = K^2 - 1$$

where $K = x^2+x-1$.

Q4. a)

$$i) y = (3+x)(1+x)^3(1-x)(3-x)^2$$



$$ii) y = \frac{x^2(x-3)}{(x-2)^2}$$

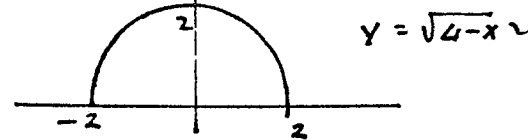
$$\begin{array}{r} x+1 \\ x^2-4x+4 \overline{) x^3-3x^2+0x+0} \\ \underline{x^3-4x^2+4x} \\ x^2-4x+4 \\ \underline{-x^2+4x-4} \\ -4 \end{array}$$

$$y = x+1 - \frac{4}{(x-2)^2}$$

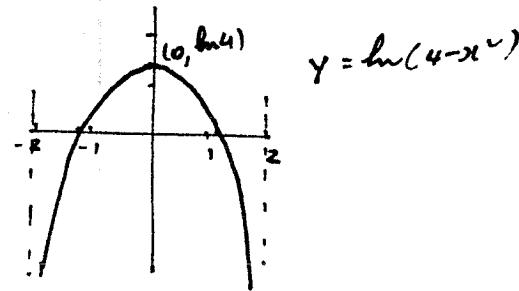
$\therefore$ Curve is always below $y = x+1$.
iii) $\rightarrow$ see end.

$$Q4 b) f(x) = 4-x^2$$

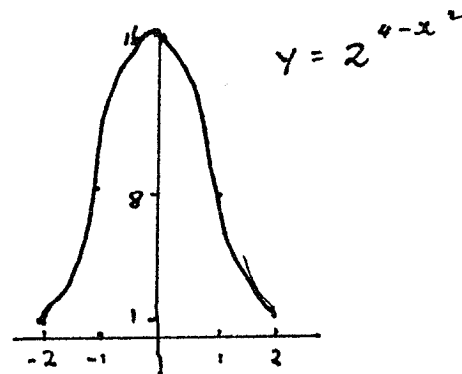
i)



ii)



iii)



$$Q5 a) \frac{x^2}{9} + \frac{y^2}{25} = 1$$

$$i) b^2 = a^2(1-e^2)$$

$$25 = 9$$

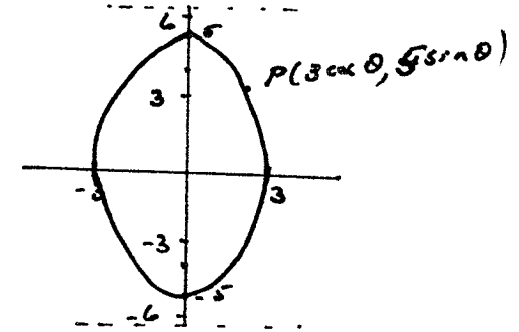
$$9 = 25(1-e^2)$$

$$\frac{9}{25} = 1-e^2, e^2 = \frac{16}{25}, e = \frac{4}{5}$$

$$ii) \text{ foci } (0, \pm 4)$$

$$\text{directrices. } y = \pm \frac{25}{4}$$

iii)



$$iv) \frac{2x}{9} + \frac{2y}{25} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{x}{9} \cdot \frac{25}{y}$$

$$\text{at } (3 \cos \theta, 5 \sin \theta)$$

$$m = \frac{-25 \times 3 \cos \theta}{3 \times 5 \sin \theta}$$

$$= -\frac{5 \cos \theta}{\sin \theta}$$

$$\therefore \text{ Tan } y - 5 \sin \theta = \frac{-5 \cos \theta}{\sin \theta} (x - 3 \cos \theta)$$

$$3y \sin \theta - 15 \sin^2 \theta = -6x \cos \theta + 15 \cos^2 \theta$$

$$\therefore 5x \cos \theta + 3y \sin \theta = 15$$

$$\text{OR } \frac{x \cos \theta}{3} + \frac{y \sin \theta}{5} = 1$$

$$v) 5x \cos \theta + 3y \sin \theta = 15$$

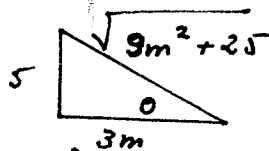
$$3y \sin \theta = -5x \cos \theta + 15$$

$$y = \frac{-5 \cos \theta \cdot x}{3 \sin \theta} + \frac{15}{3 \sin \theta}$$

$$= -\frac{5x}{3} \cot \theta + 5 \operatorname{cosec} \theta$$

$$\text{Let } m = -\frac{5 \cot \theta}{3}$$

$$\therefore \frac{3m}{5} = \cot \theta$$



$$\therefore y = mx \pm \frac{5 \cdot \sqrt{9m^2 + 25}}{5}$$

$$= mx \pm \sqrt{9m^2 + 25}$$

$$vi) \text{ If } m = 2$$

$$y = 2x \pm \sqrt{61}$$

$$b) \frac{x^2}{8} - \frac{y^2}{4} = 1 \quad [x^2 - 2y^2 = 8]$$

$$\frac{2x}{8} - \frac{2y}{4} \cdot \frac{dy}{dx} = 0$$

$$\therefore -\frac{y}{4} \cdot \frac{dy}{dx} = -\frac{x}{8}$$

$$\therefore \frac{dy}{dx} = \frac{x}{2y}$$

$$\text{at } (x_1, y_1) \quad \frac{dy}{dx} = \frac{x_1}{2y_1}$$

$$\therefore T \text{ is } y - y_1 = \frac{x_1}{2y_1} (x - x_1)$$

$$\therefore 2y_1 y - 2y_1^2 = x_1(x - x_1)$$

$$\therefore x_1^2 - 2y_1^2 = x_1(x - 2y_1 y)$$

$$\therefore 8 = x_1(x - 2y_1 y)$$

ii) eq of C from (x_0, y_0) is

$$x_0 x - 2y_0 y = 8$$

$$\text{Now } 2x - 3y = 5$$

$$\therefore \frac{8}{5} \cdot 2x - \frac{8}{5} \cdot 3y = \frac{8}{5} \cdot 5$$

$$\therefore \frac{16x}{5} - \frac{24y}{5} = 8$$

$$\therefore x_0 = \frac{16}{5}, y_0 = -\frac{24}{5}$$

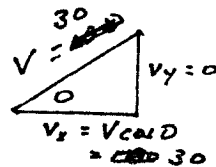
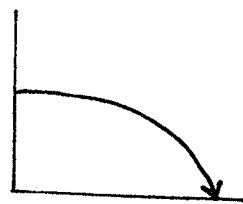
$$6a) i) \tan 75 = \tan(45 + 30)$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \cdot \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{3 + 2\sqrt{3} + 1}{2}$$

$$= 2 + \sqrt{3}$$



$$\ddot{x} = 0$$

$$\ddot{y} = -10$$

$$\dot{x} = c_1$$

$$\dot{y} = c_2 - 10t$$

$$\text{at } t = 0, \dot{x} = 30 = c_1, \dot{y} = 0 = c_2$$

$$\therefore \dot{x} = 30 \quad \dot{y} = -10t$$

$$x = 30t + c_3, y = c_4 - 5t^2$$

$$\text{at } t = 0, x = 0 = c_3, y = H = c_4$$

$$\therefore x = 30t, y = H - 5t^2$$

at any time,

$$\tan \theta = \frac{y'}{x'}$$

$$\therefore \tan 75 = \left| \frac{-10t}{30} \right|$$

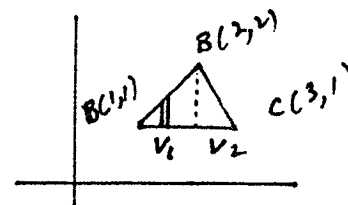
$$\therefore 2\sqrt{3} = \frac{t}{3}$$

$$t = 3(2\sqrt{3}) \text{ sec}$$

Hits ground at $y = 0$

$$\therefore H = 5 \times 3(2\sqrt{3}) \text{ m} = 15(2\sqrt{3}) \text{ m}$$

b)



Line AB is $y = x$

Line BC is $y = 4 - x$

$$V_{\text{shell}} = \pi R^2 H - \pi r^2 h$$

$$V_i: R = x + dx, r = x, H = y - 1 = h$$

$$\therefore V_{\text{shell}} = \pi (y-1) \{ (x+dx)^2 - x^2 \}$$

$$= \pi (y-1) (x^2 + 2x dx + dx^2 - x^2)$$

$$= \pi (y-1) (2x dx), \quad dx^2 \approx 0$$

$$V_{\text{solid}} = \pi \int_1^2 (x-1) 2x \cdot dx$$

$$= 2\pi \int_1^2 (x^2 - x) dx$$

$$= 2\pi \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2$$

$$= 2\pi \left(\frac{8}{3} - \frac{4}{2} - \left(\frac{1}{3} - \frac{1}{2} \right) \right) = 2\pi \left(\frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{2} \right) = 2\pi \left(\frac{16}{6} - \frac{12}{6} - \frac{2}{6} + \frac{3}{6} \right) = 2\pi \left(\frac{3}{6} \right) = \frac{2\pi}{3}$$

$$8c) \int_0^4 \sqrt{2-\sqrt{x}} \, dx.$$

$$\text{let } x = 4 \sin^4 \theta$$

$$\therefore dx = 16 \sin^3 \theta \cdot \cos \theta \cdot d\theta.$$

$$\text{at } x=0, \theta=0$$

$$x=4, \theta = \pi/2$$

$$\therefore I = \int_0^{\pi/2} \sqrt{2-2\sin^4 \theta} \cdot 16 \sin^3 \theta \cos \theta \cdot d\theta$$

$$= \int_0^{\pi/2} \sqrt{2} \sqrt{1-\sin^4 \theta} \cdot 16 \sin^3 \theta \cos \theta \cdot d\theta$$

$$= \sqrt{2} \int_0^{\pi/2} \cos \theta \cdot 16 \sin^3 \theta \cdot \cos \theta \cdot d\theta$$

$$= 16\sqrt{2} \int_0^{\pi/2} \sin^3 \theta \cos^2 \theta \cdot d\theta$$

$$= 16\sqrt{2} \int_0^{\pi/2} \sin \theta \cdot \cos^4 \theta \cdot (1-\cos^2 \theta) d\theta$$

$$= 16\sqrt{2} \int_0^{\pi/2} \cos^4 \theta \sin \theta \cdot d\theta$$

$$- 16\sqrt{2} \int_0^{\pi/2} \cos^6 \theta \sin \theta \cdot d\theta$$

$$= 16\sqrt{2} \left\{ \left[\frac{1}{3} \cos^3 \theta - \frac{1}{5} \cos^5 \theta \right]_0^{\pi/2} \right\}$$

$$= 16\sqrt{2} \left\{ 0 - \left(\frac{1}{3} - \frac{1}{5} \right) \right\}$$

$$= 16\sqrt{2} \cdot \frac{2}{15} = \frac{32\sqrt{2}}{15}$$

$$4a) iii) 4y^2 = x^2 - 4x.$$

$$x^2 - 4x - 4y^2 = 0$$

$$x^2 - 4x + 4 - 4y^2 = 4$$

$$(x-2)^2 - 4y^2 = 4$$

$$\therefore \frac{(x-2)^2}{4} - \frac{y^2}{1} = 1$$

Hyperbola, centre at (2,0)

$$\text{as } y \rightarrow \infty \quad 4y^2 \rightarrow (x-2)^2 - 4$$

$$4y^2 \rightarrow (x-2)^2$$

$$2y \rightarrow \pm (x-2)$$

$$y \rightarrow \pm \frac{x-2}{2}$$

$\therefore$ Asymptotes are

$$y = \frac{x}{2} - 1, \quad -\frac{x}{2} + 1$$

